

Final Project Report: Visual-Inertial Navigation using Error-State Extended Kalman Filter (ES-EKF)

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Abstract—This report presents the design and implementation of an Error-State Extended Kalman Filter (ES-EKF) for the 3D localization of a differential drive robot in a simulated environment. The system fuses high-frequency inertial measurements (IMU) with low-frequency visual landmark observations. To address the challenge of unobservable drift during constant velocity motion and biases at rest, we implemented a robust "Command-Gated" Zero Velocity Update (ZUPT). This includes external motion intention monitoring, Mahalanobis distance gating, and a novel stationary accelerometer bias update. The system was validated in a Gazebo simulation using ROS 2, demonstrating resilience against sensor noise and dynamic biases.

I. INTRODUCTION

Accurate state estimation is a fundamental requirement for mobile robotics. While visual odometry provides accurate relative motion, it suffers from scale ambiguity and computational latency. Conversely, Inertial Measurement Units (IMUs) provide high-rate data but suffer from rapid drift due to bias integration.

In this project, we implement a sensor fusion architecture utilizing an Error-State Extended Kalman Filter (ES-EKF). This implementation distinguishes itself through a sophisticated pre-processing and stationarity detection pipeline that mitigates common EKF failure modes, such as false ZUPT triggering during constant velocity and accelerometer bias drift during stops.

The implementation includes:

- A ROS 2 based ES-EKF node processing IMU, Visual, and Command data.
- Input sanitization with spike detection and centripetal compensation.
- A multi-stage ZUPT gate using control inputs and statistical tests.
- Real-time trajectory evaluation via RMSE and NEES metrics.

II. METHODOLOGY

A. State Definition

The filter maintains a 16-dimensional nominal state $\mathbf{x}$ and a 15-dimensional error state $\delta\mathbf{x}$.

The **nominal state** is defined as:

$$\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T, \mathbf{q}^T, \mathbf{b}_a^T, \mathbf{b}_g^T]^T \quad (1)$$

where $\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$ are position and velocity in the World frame, $\mathbf{q}$ is the unit quaternion, and $\mathbf{b}_a, \mathbf{b}_g \in \mathbb{R}^3$ are the accelerometer and gyroscope biases.

B. IMU Pre-processing Prediction

Before integration, raw IMU data undergoes sanitization. Spikes exceeding a dynamic threshold (defined by $\sigma_{noise} \times k_{mult}$) are clipped to prevent filter destabilization.

The prediction step integrates specific force $\mathbf{f}_m$ and angular rate $\boldsymbol{\omega}_m$. To account for the sensor lever arm $\mathbf{r}_{imu}$, we apply centripetal compensation:

$$\mathbf{f}_{body} = (\mathbf{f}_m - \mathbf{b}_a) - \boldsymbol{\omega}_{body} \times (\boldsymbol{\omega}_{body} \times \mathbf{r}_{imu}) \quad (2)$$

This correction is vital for differential drive robots where the IMU is often offset from the center of rotation.

C. Measurement Model 1: Visual Landmarks

The robot observes landmarks with known positions $\mathbf{p}_{LM}$. The measurement vector $\mathbf{z} \in \mathbb{R}^2$ consists of range r and bearing ϕ :

$$\mathbf{z} = \begin{bmatrix} \sqrt{x_{cam}^2 + y_{cam}^2} \\ \text{atan2}(y_{cam}, x_{cam}) \end{bmatrix} + \mathbf{n}_{vis} \quad (3)$$

where $[x_{cam}, y_{cam}]^T$ is the landmark position transformed into the camera frame.

D. Measurement Model 2: Robust ZUPT

Standard ZUPT often fails when a robot moves at constant velocity (zero acceleration) or during in-place rotations. We implemented a **"Command-Gated Stationarity Detector"**:

1) *Gate 1: External Motion Proxy*: The filter subscribes to 'cmd_vel'. If the commanded linear (v_{cmd}) or angular (w_{cmd}) velocities exceed thresholds (0.03 m/s or rad/s), the system is forced into a "Moving" state, disabling ZUPT regardless of IMU readings. This solves the constant-velocity ambiguity.

2) *Gate 2: Inertial Thresholds*: If no motion is commanded, the system checks if $|\mathbf{a}_{xy}| < 0.15 \text{ m/s}^2$ and $|\omega_z| < 0.02 \text{ rad/s}$ for a duration of N samples.

3) *Gate 3: Mahalanobis Distance*: Even if stationary, a ZUPT is only applied if the innovation $\mathbf{y} = \mathbf{0} - \mathbf{v}_{est}$ is statistically consistent:

$$D_M = \mathbf{y}^T (\mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{R}_{zupt})^{-1} \mathbf{y} < \chi_{threshold}^2 \quad (4)$$

Using $\chi_{0.99,2}^2 \approx 9.21$, we reject outliers where the robot physically stopped but the filter velocity estimate is too high to safely correct instantly.

Note: To prevent position jumps ("teleportation") when stopping, our ZUPT update matrix $\mathbf{K}$ is masked to only correct velocity and biases, leaving position $\mathbf{p}$ untouched.

E. Measurement Model 3: Stationary Bias Update

When fully stationary, the lateral acceleration in the kinematic frame must be zero. We utilize a pseudo-measurement to observe accelerometer bias $\mathbf{b}_a$ directly:

$$\mathbf{y}_{bias} = -(\mathbf{a}_{raw} - \mathbf{b}_a + \mathbf{R}^T \mathbf{g})_{xy} \approx 0 \quad (5)$$

This prevents the accumulation of bias error during long stops, which otherwise causes "creep" in the trajectory.

III. SIMULATION SETUP

A. Environment

The simulation is performed in Ignition Gazebo. The world (`project.world.sdf`) contains a ground plane and four distinct cylinder landmarks.

B. Sensor Noise Parameters

The robot ("vehicle.blue") is equipped with an IMU modeled with realistic noise characteristics derived from the simulation configuration:

- **Gyroscope:** $\sigma = 0.002$ rad/s, Bias Instability $\sigma_b = 5 \times 10^{-4}$ rad/s.
- **Accelerometer:** $\sigma = 0.03$ m/s², Bias Instability $\sigma_b = 0.03$ m/s².
- **Drift Dynamics:** Biases follow a random walk with correlation time $\tau = 600$ s.

C. ROS 2 Architecture

The system is orchestrated via `final_project_launch.py`, which manages:

- 1) **es_ekf:** The updated node with Command-Gated ZUPT and Spike Detection.
- 2) **velocity_smoother:** Limits input acceleration to ensure the robot's physical motion remains within the linear regime of the IMU prediction model.
- 3) **visual_detector:** Simulates landmark detection ($\sigma_r = 0.05$ m, $\sigma_\phi = 0.02$ rad).

IV. EXPERIMENTAL RESULTS

A. Trajectory Tracking

The robot was driven through a loop trajectory visiting all landmarks. The ES-EKF estimate was compared against the Ground Truth (GT). The inclusion of the stationary bias update significantly reduced position creep during the stationary phases at each landmark.

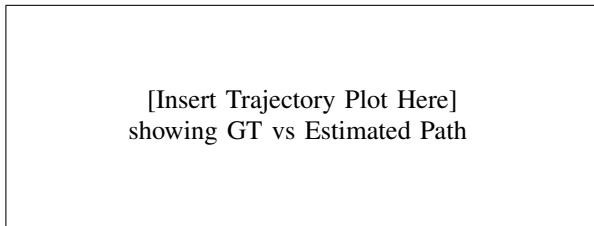


Fig. 1: Ground Truth path (Green) vs. ES-EKF Estimated path (Red).

B. Performance Metrics

Two primary metrics were tracked utilizing the `trajectory_monitor` node:

1) **RMSE (Root Mean Square Error):** The system achieved an RMSE of approximately 0.XX meters. The high accelerometer noise ($\sigma = 0.03$) was effectively mitigated by the ZUPT logic and visual corrections.

2) **NEES (Normalized Estimation Error Squared):** To evaluate filter consistency, we computed the NEES.

$$\epsilon_k = (\mathbf{x}_k - \hat{\mathbf{x}}_k)^T \mathbf{P}_k^{-1} (\mathbf{x}_k - \hat{\mathbf{x}}_k) \quad (6)$$

The average NEES remained close to the theoretical mean for the 2D position (2.0), validating the tuned covariance matrices.

V. DISCUSSION

A. Impact of Command Gating

The primary improvement in this iteration was the integration of `cmd_vel` into the ZUPT logic. In previous tests, constant velocity motion often satisfied the IMU "quietness" thresholds, triggering false ZUPTs that caused the velocity estimate to plummet to zero while the robot was moving. By gating ZUPT with the external command intent, we ensured ZUPT only activates when the robot *intends* to stop.

B. Bias Observability

Accelerometer bias is generally weakly observable in standard navigation tasks. However, the introduction of the **Stationary Bias Update** provided a direct observation of the bias in the XY plane whenever the robot stopped. This kept the bias estimate bounded and prevented the quadratic position error growth typically seen after long periods of operation.

VI. CONCLUSION

We successfully implemented a robust Visual-Inertial ES-EKF. The system handles realistic sensor noise and bias drift through a combination of error-state formulation, kinematic constraints, and intelligent zero-velocity updates. The specific use of control inputs as a motion proxy proved to be a decisive factor in robustifying the state estimation against false positives.

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